CS 412 / 512
Introduction to Computer Graphics

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COMPUTER SCIENCE

Who am I

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Charles Omaoeng

Office Hour:
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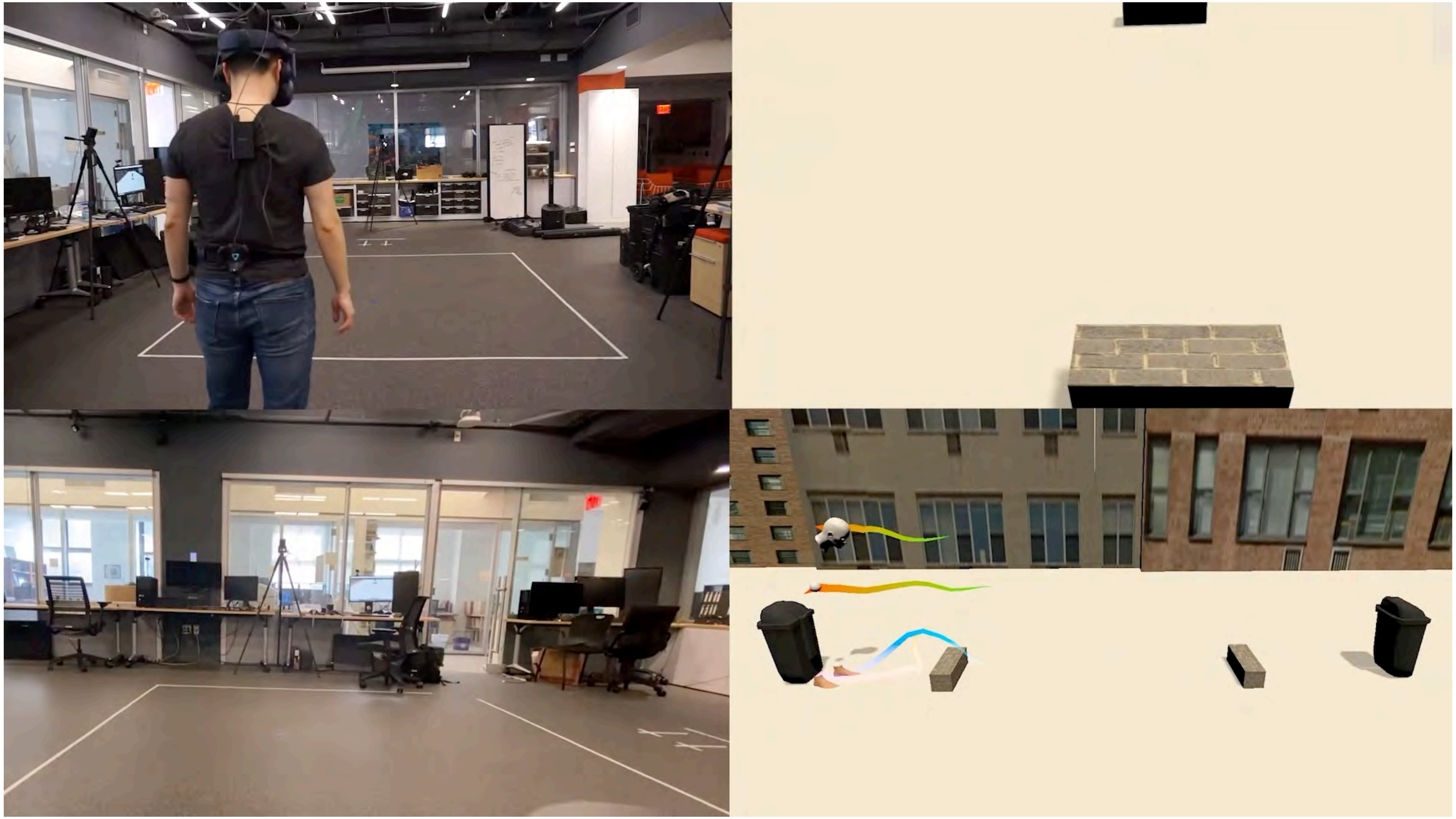
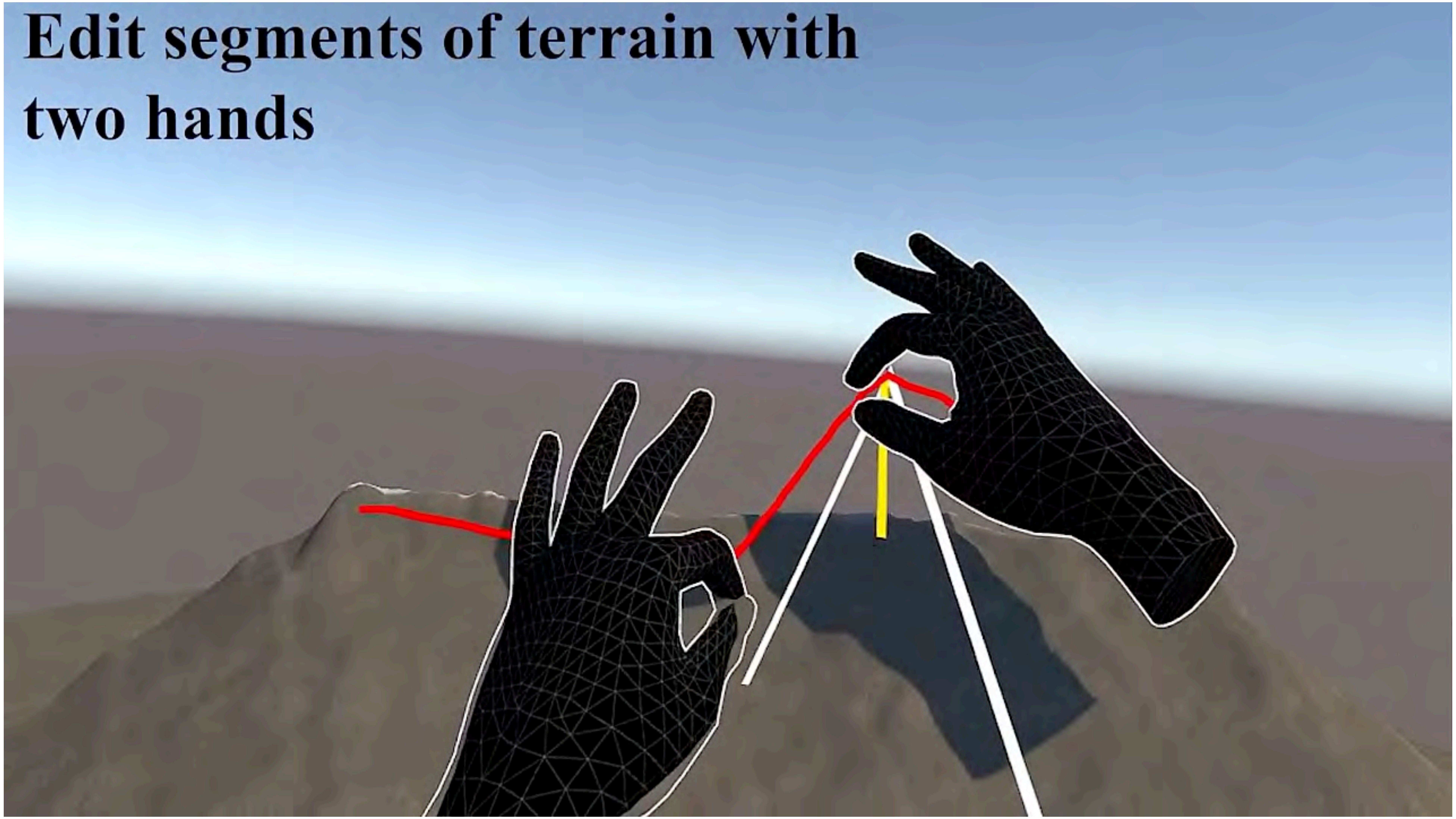
What I do

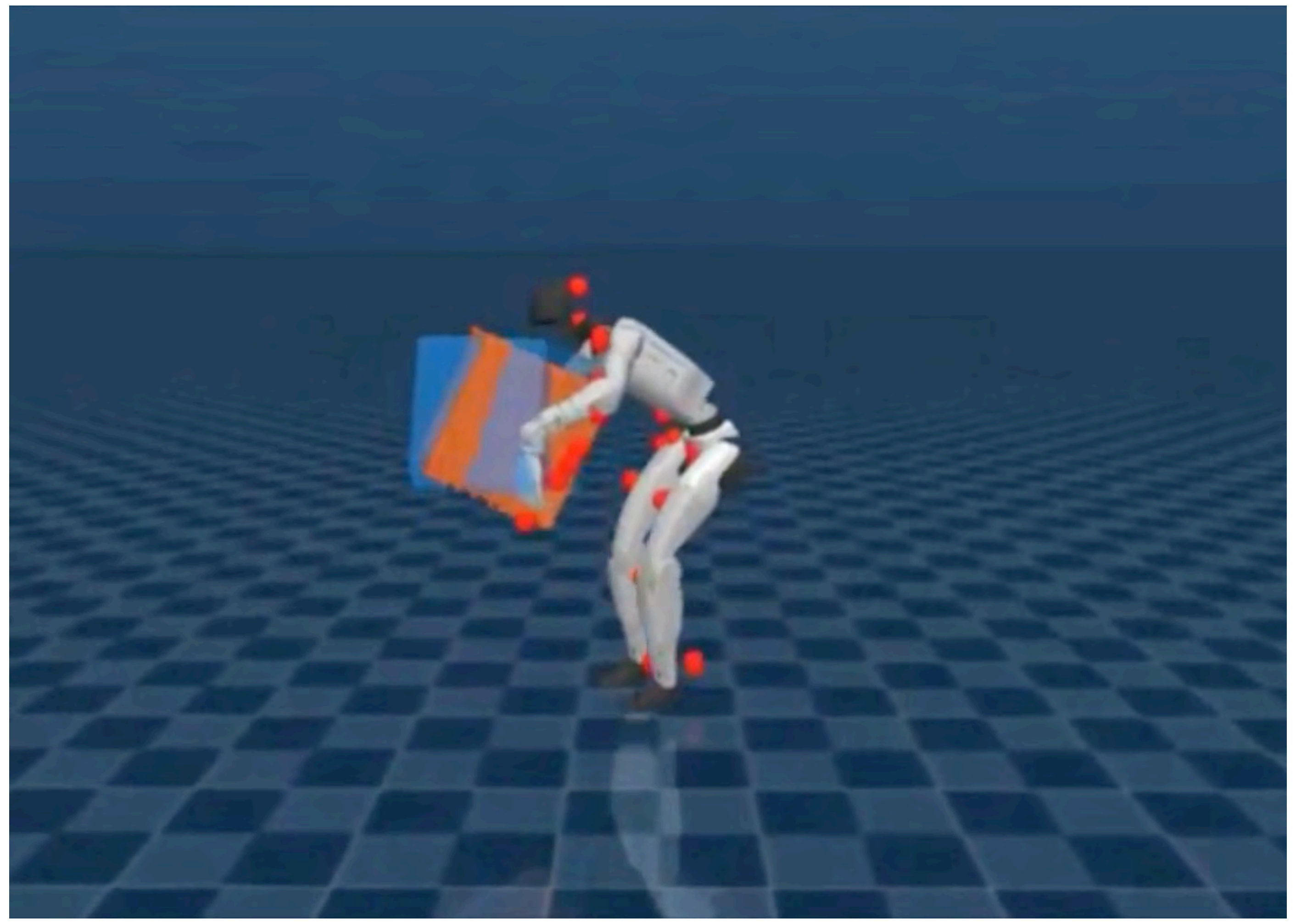


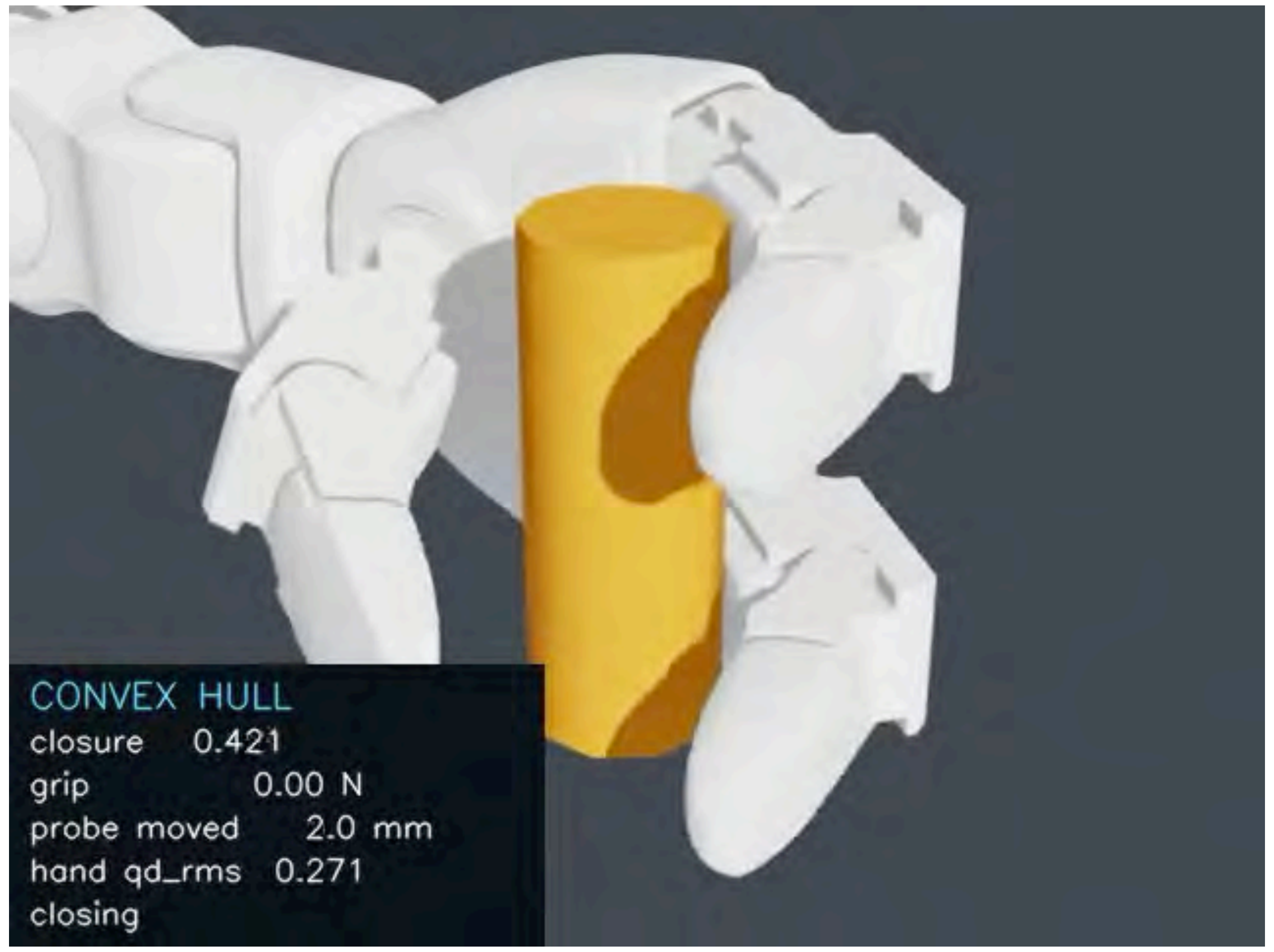

Edit segments of terrain with two hands

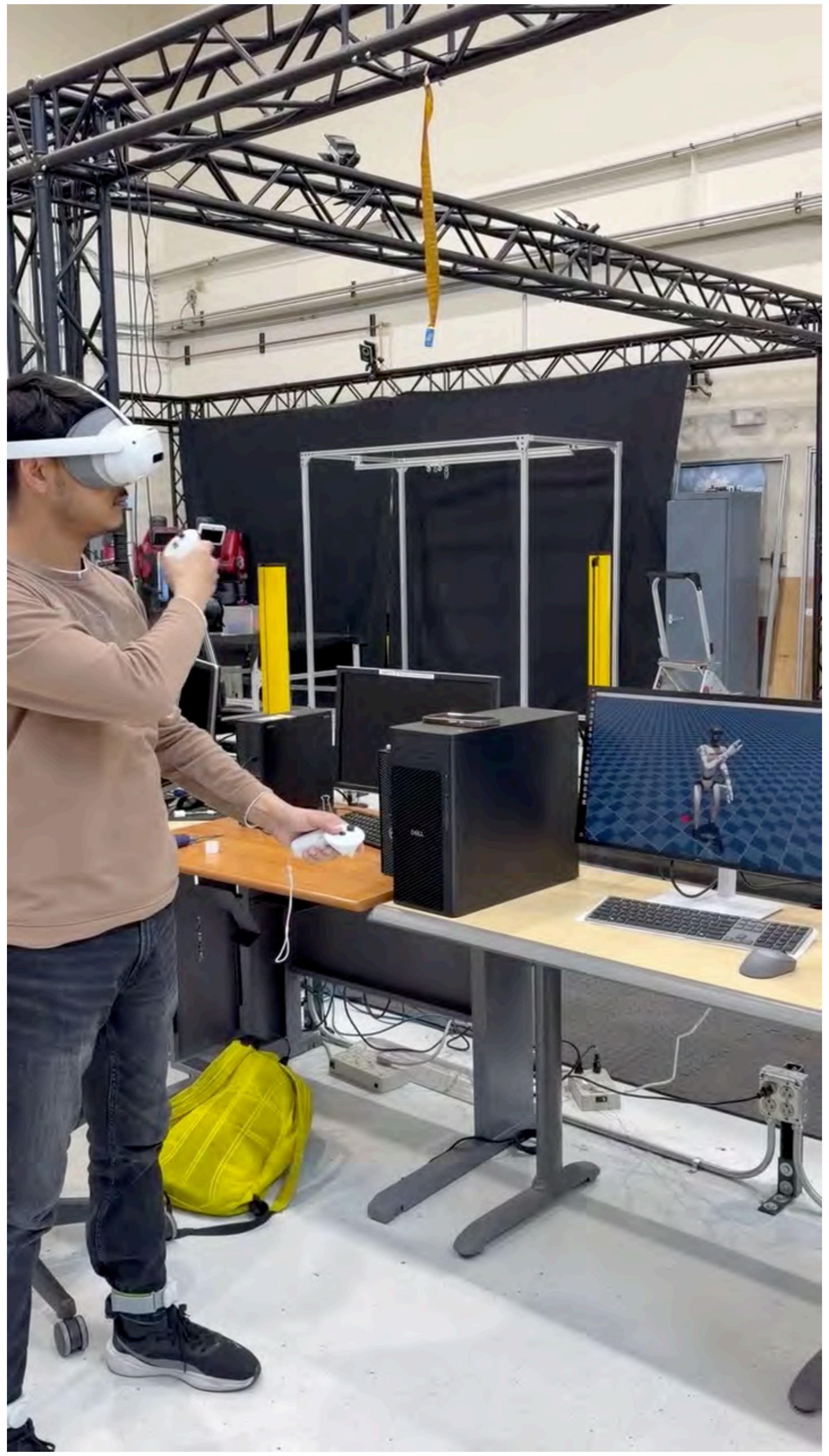
Selected Latest Work

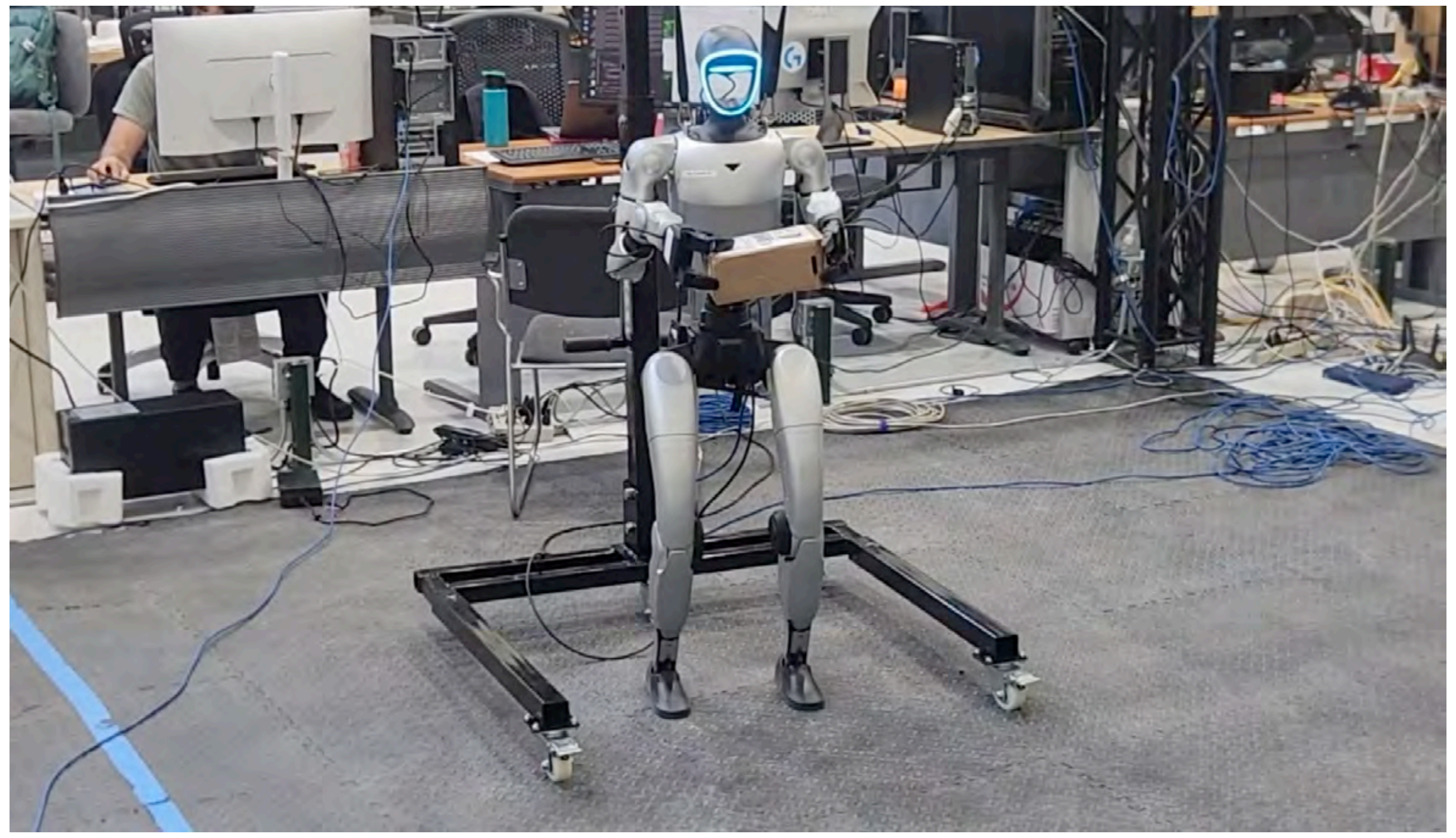
Unpublished research, not for distribution

Selected Latest Work





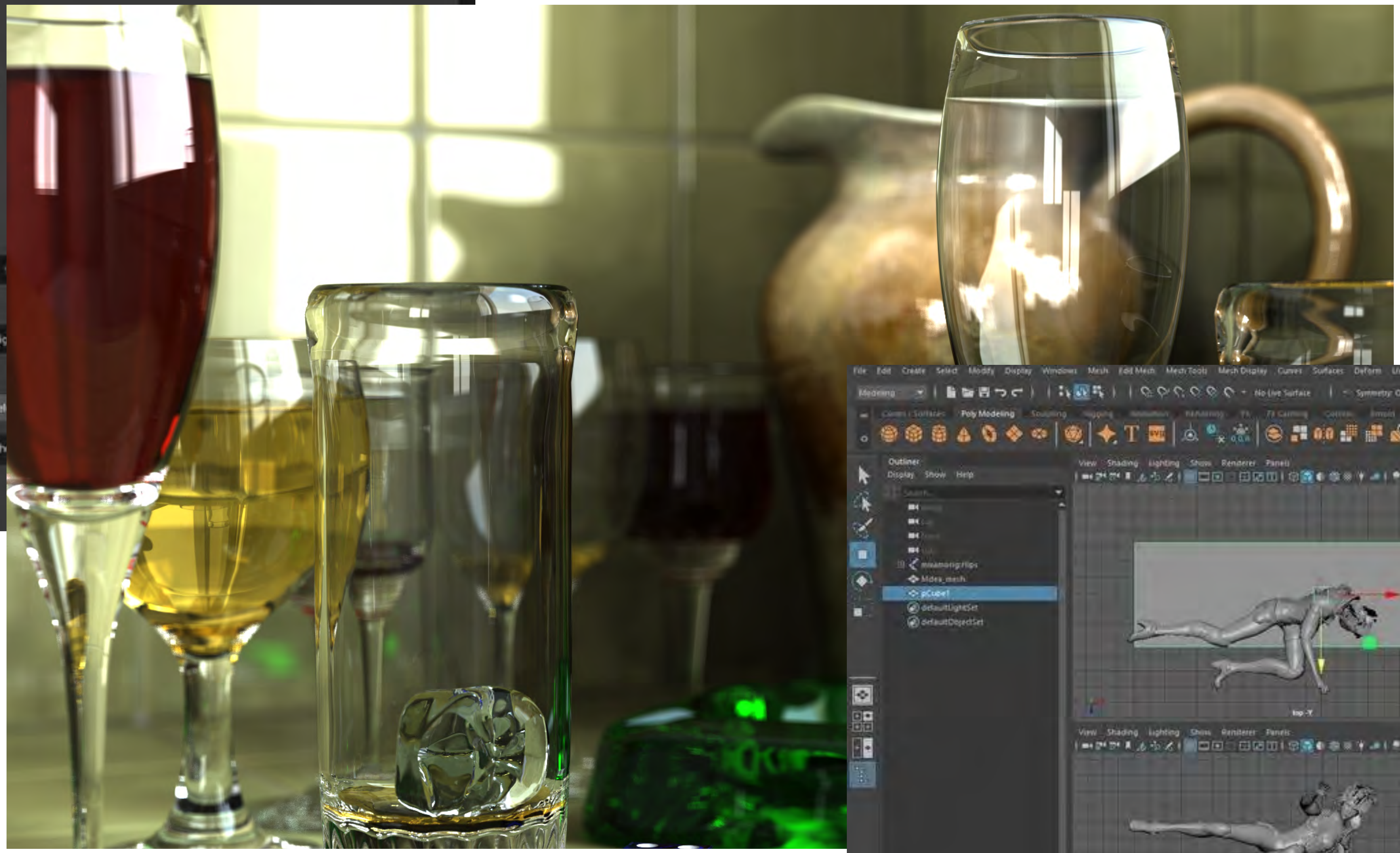




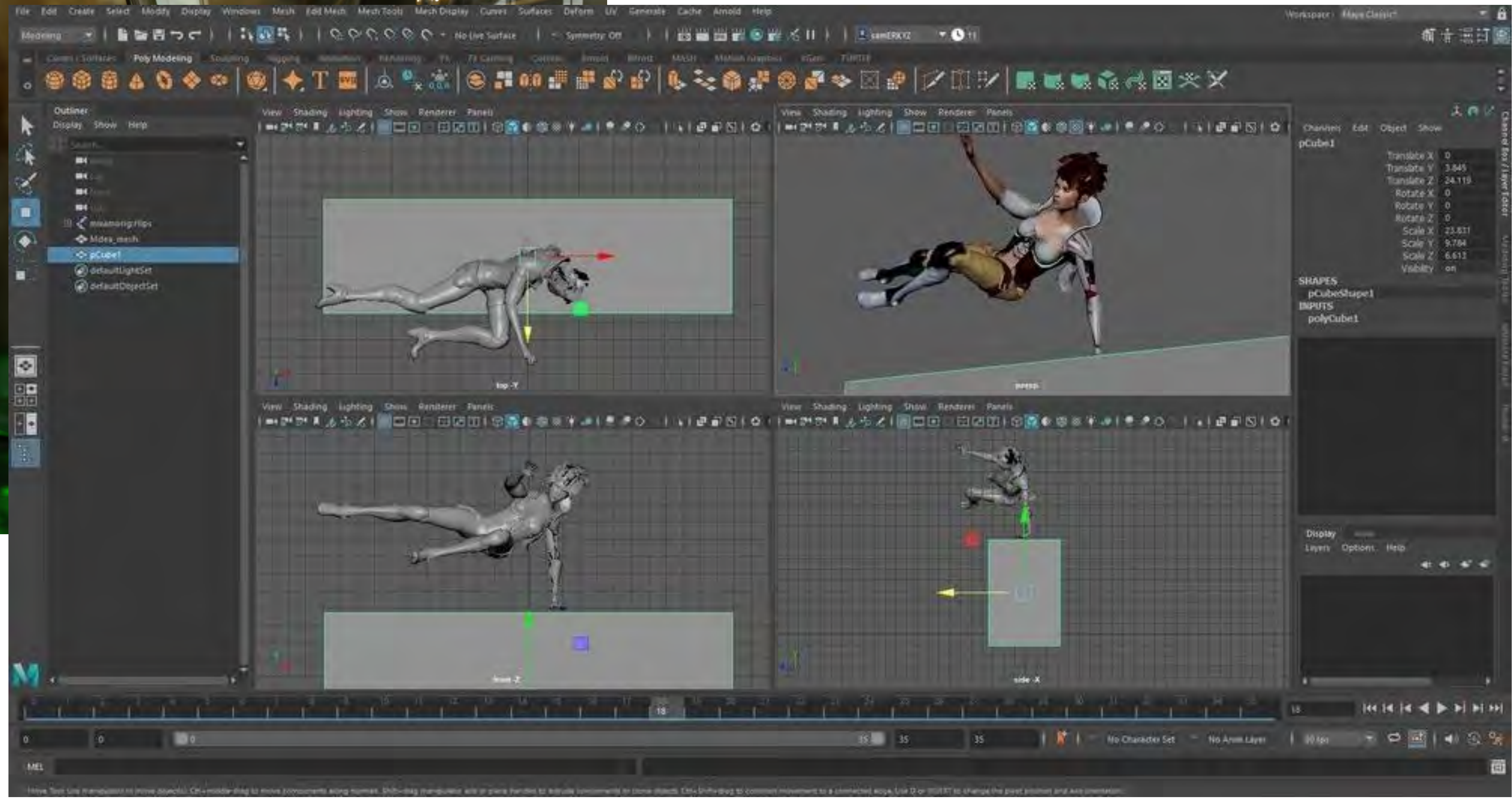
Unpublished research, not for distribution

Why learn it?

[source: Blender]

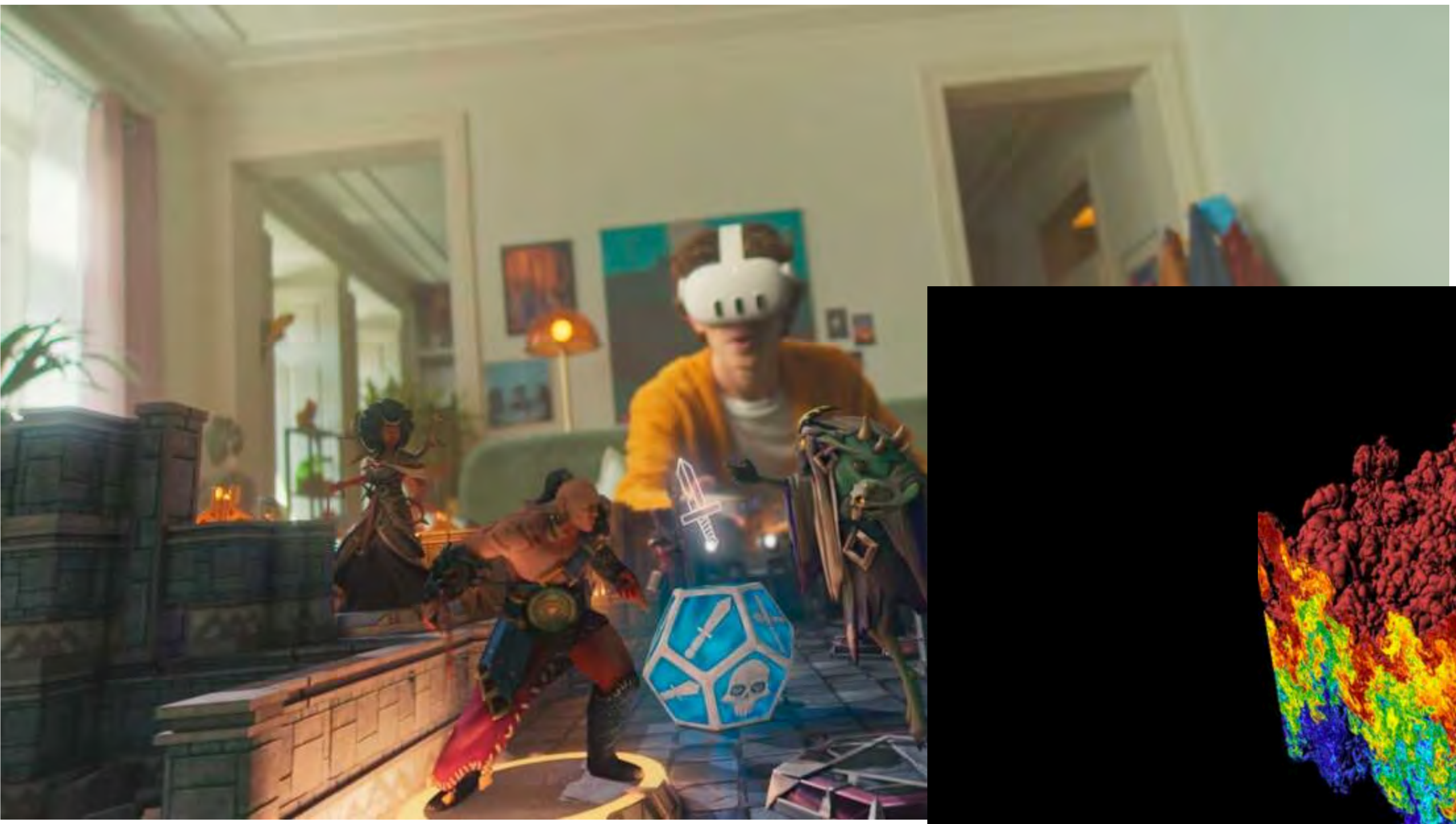


[source: Wikipedia]

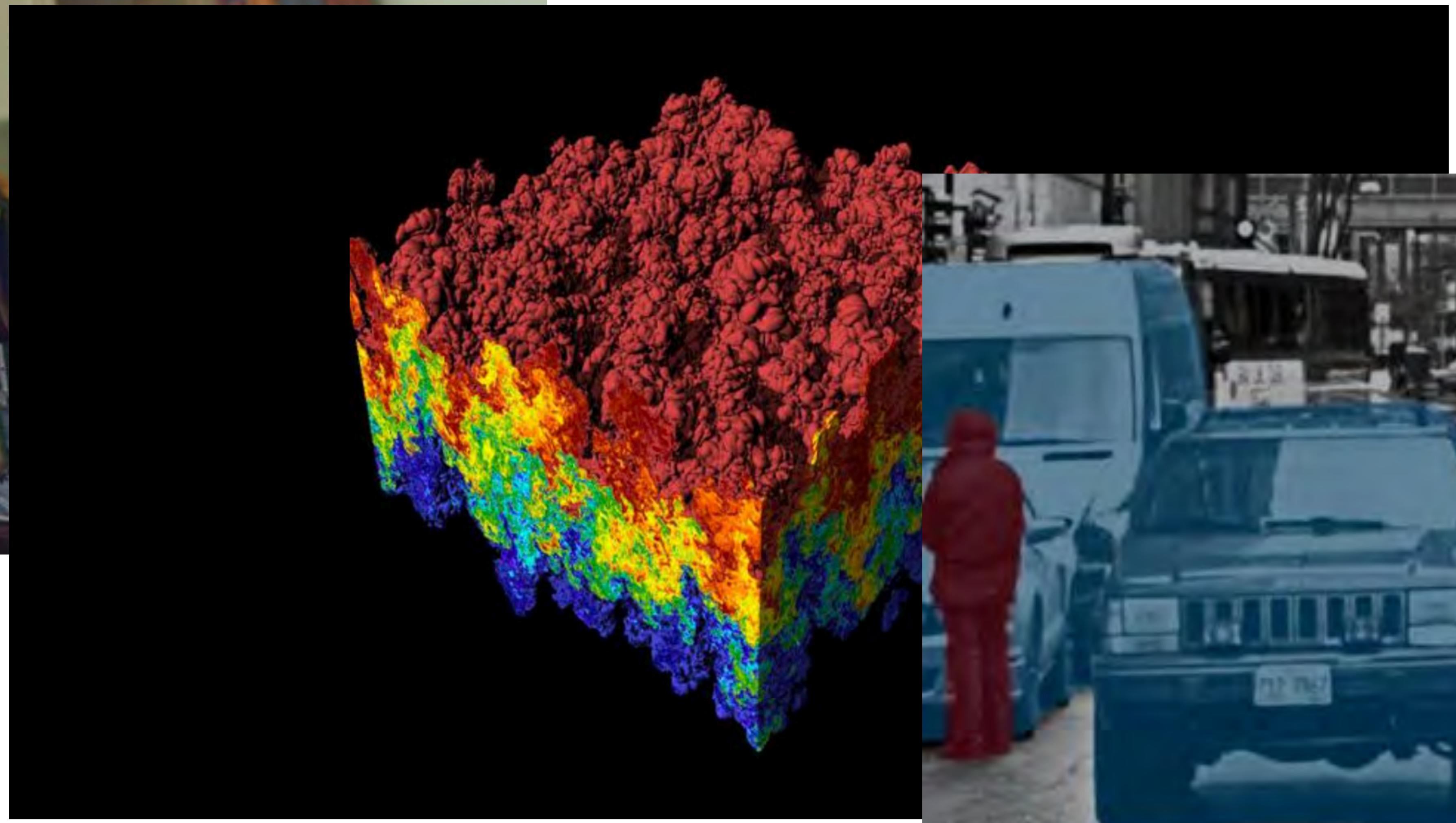


[source: Maya]

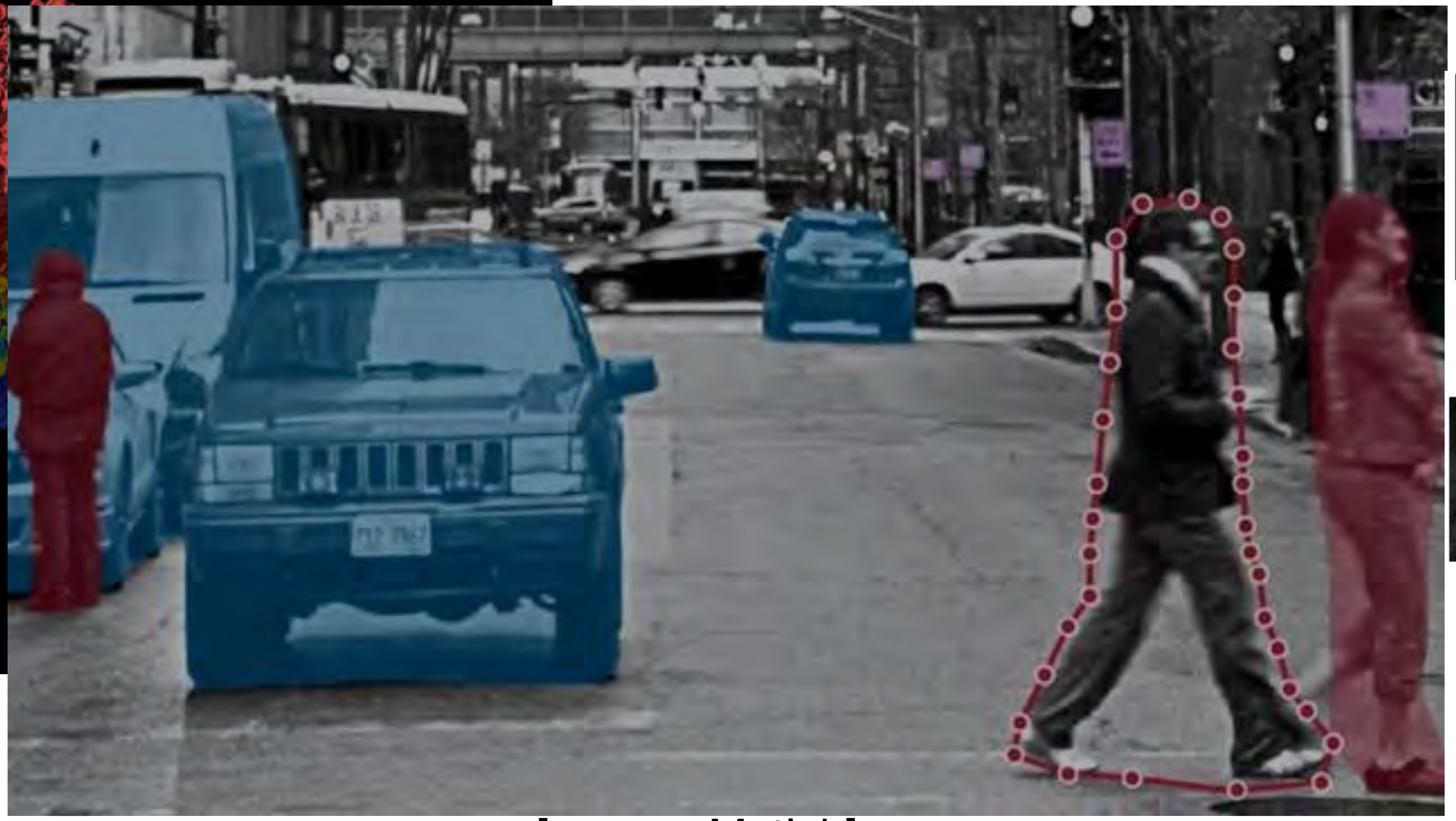
Why learn it?



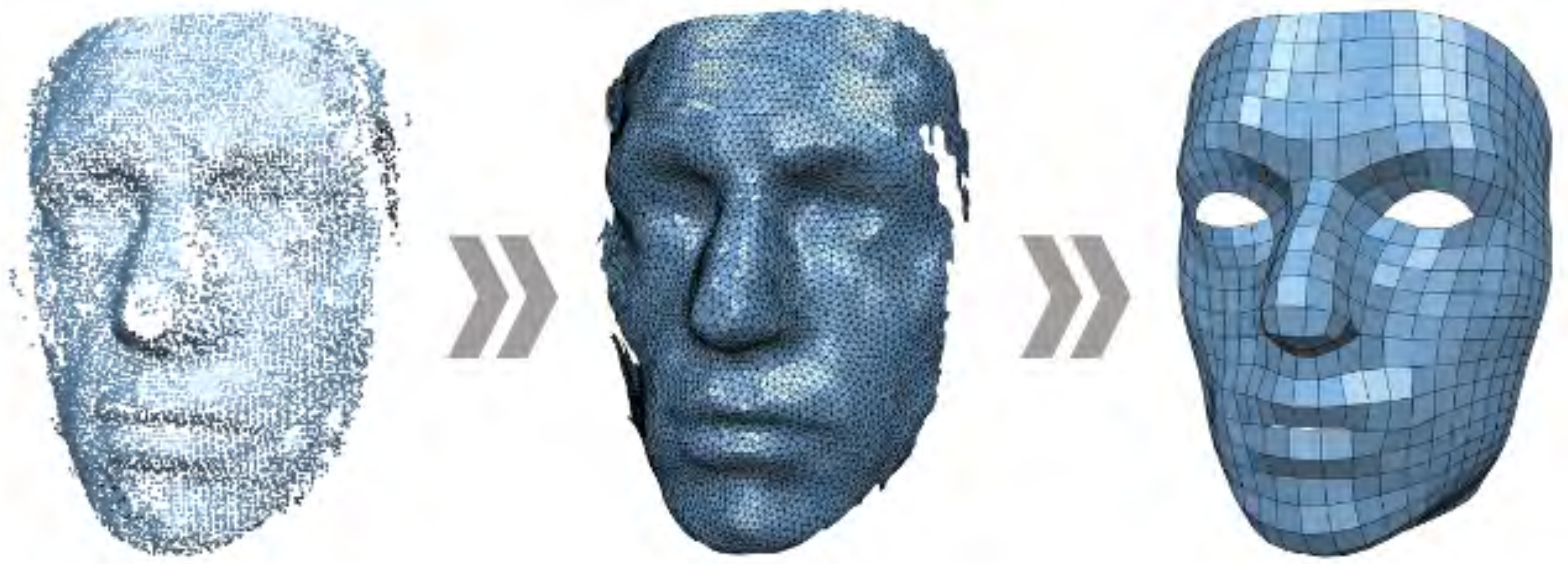
[source: Meta]



[source: Wikipedia]



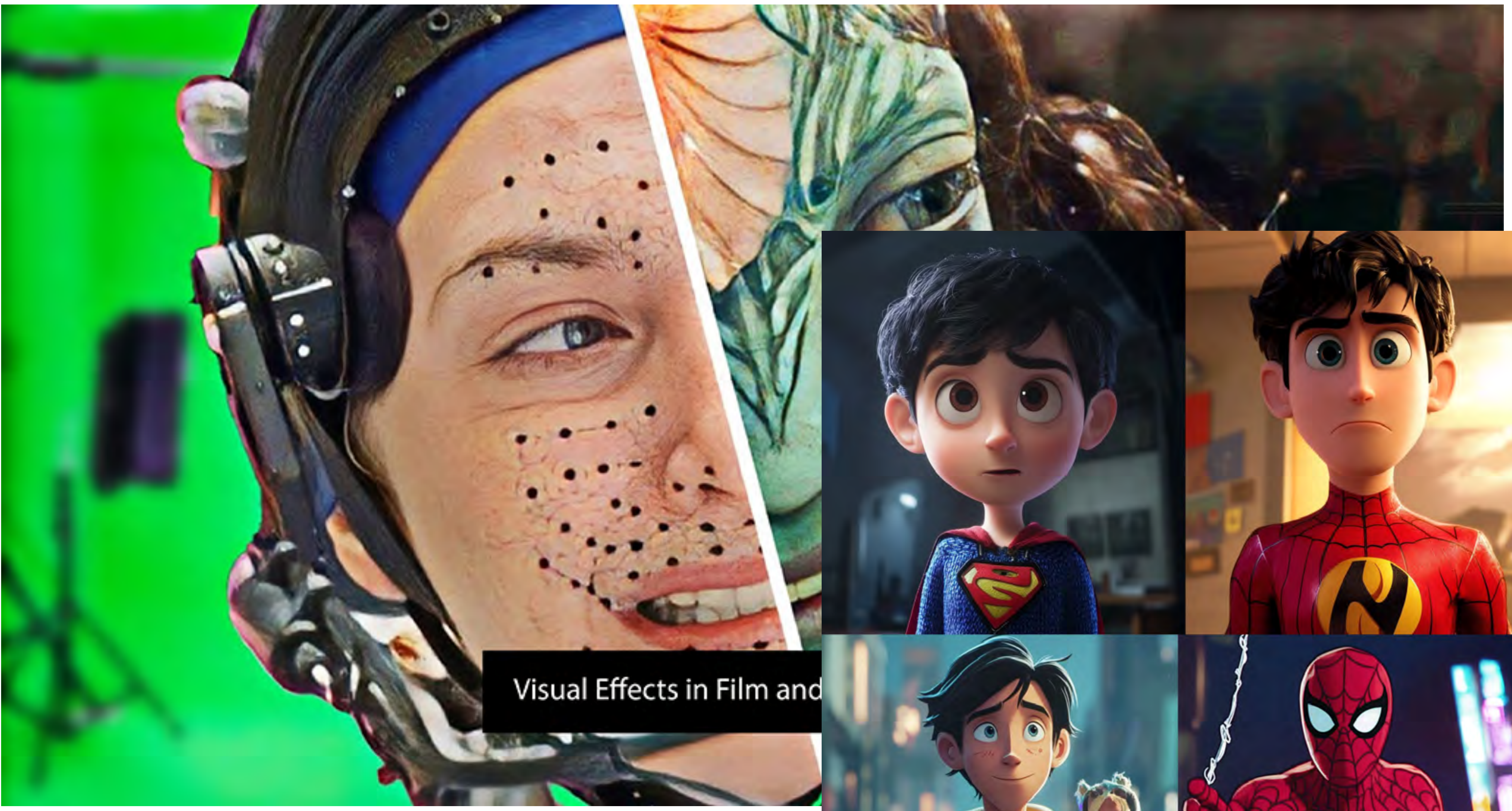
[source: Matlab]



[source: Rwth AAchen]

[source: Wikipedia]

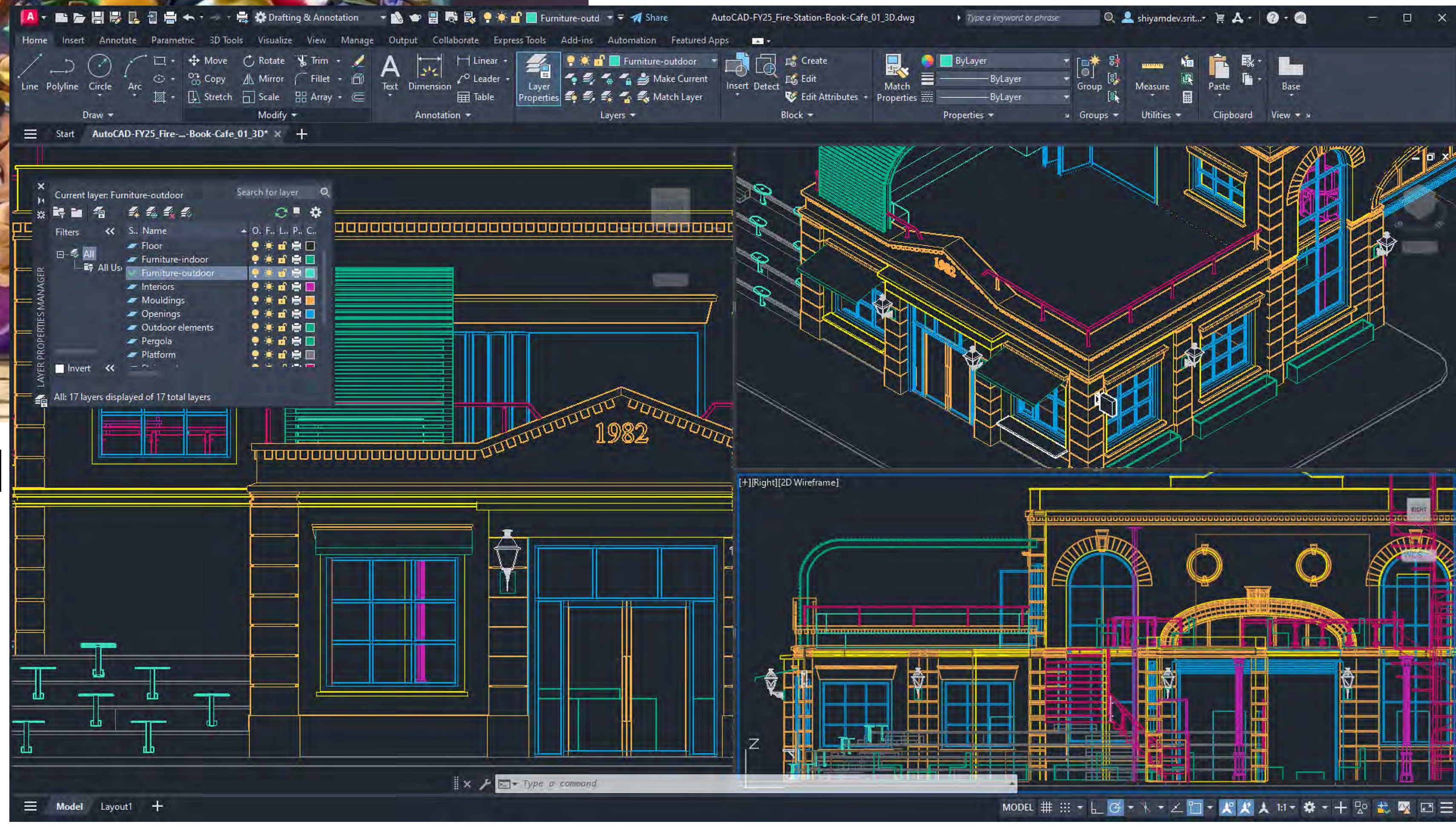
Applications



[source: Avatar]

[source: Toons Mag]

[source: Nintendo]



[source: AudoCAD]

What is Computer Graphics

It is a field of study that explores the principles, algorithms, and systems for generating, manipulating, and displaying visual content with computers.

- Modeling (what it is)
- Rendering (how it looks)
- Animation (how it changes over time)

Learning Objectives

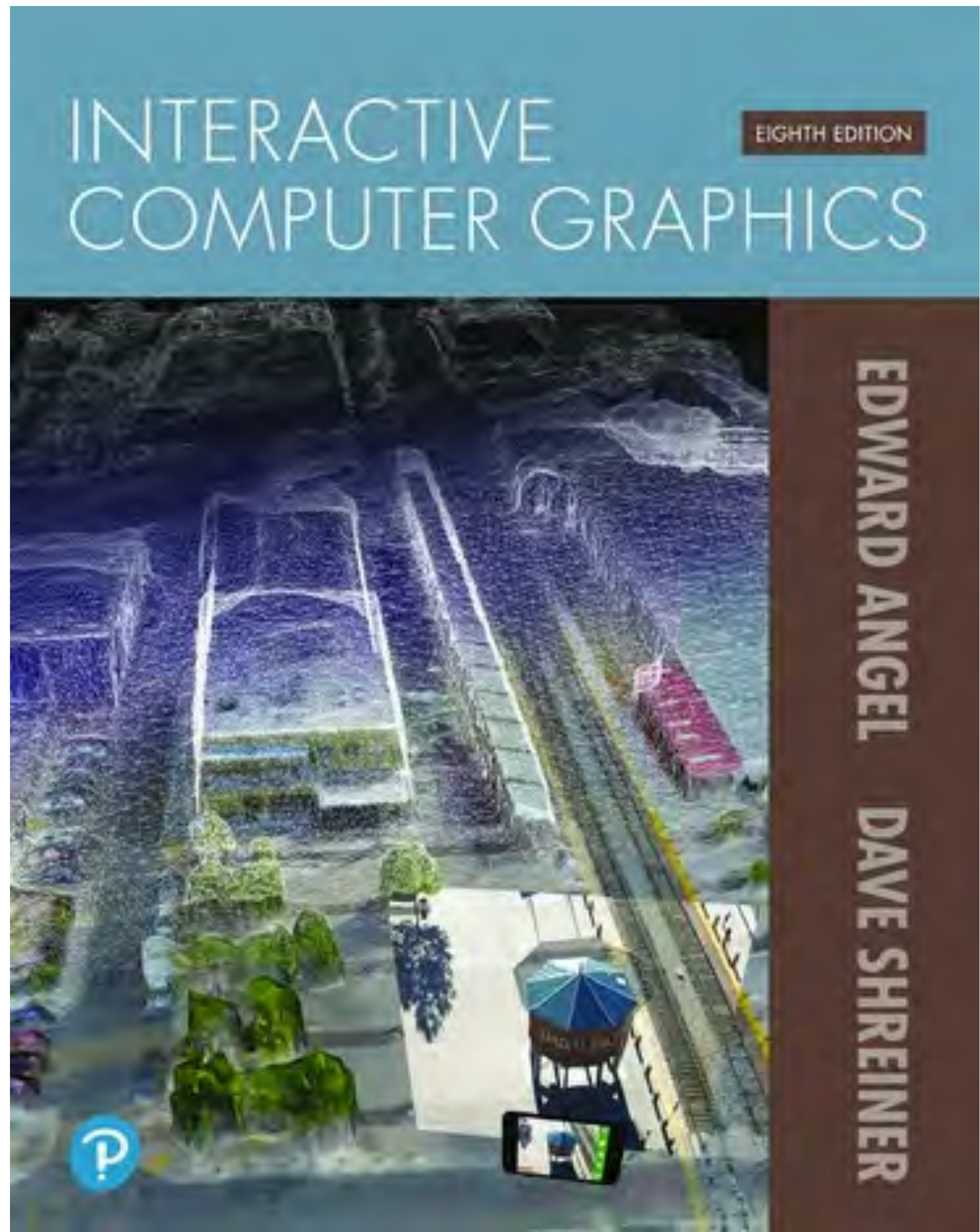
- Mathematical foundations
- Core modeling, rendering, animation
- Shaders (parallel programs that run on GPU)
- Implementing a rendering pipeline with WebGL and JavaScript
- (Stretch goal, optional) *Some advanced rendering algorithms*

This course DOES NOT cover:

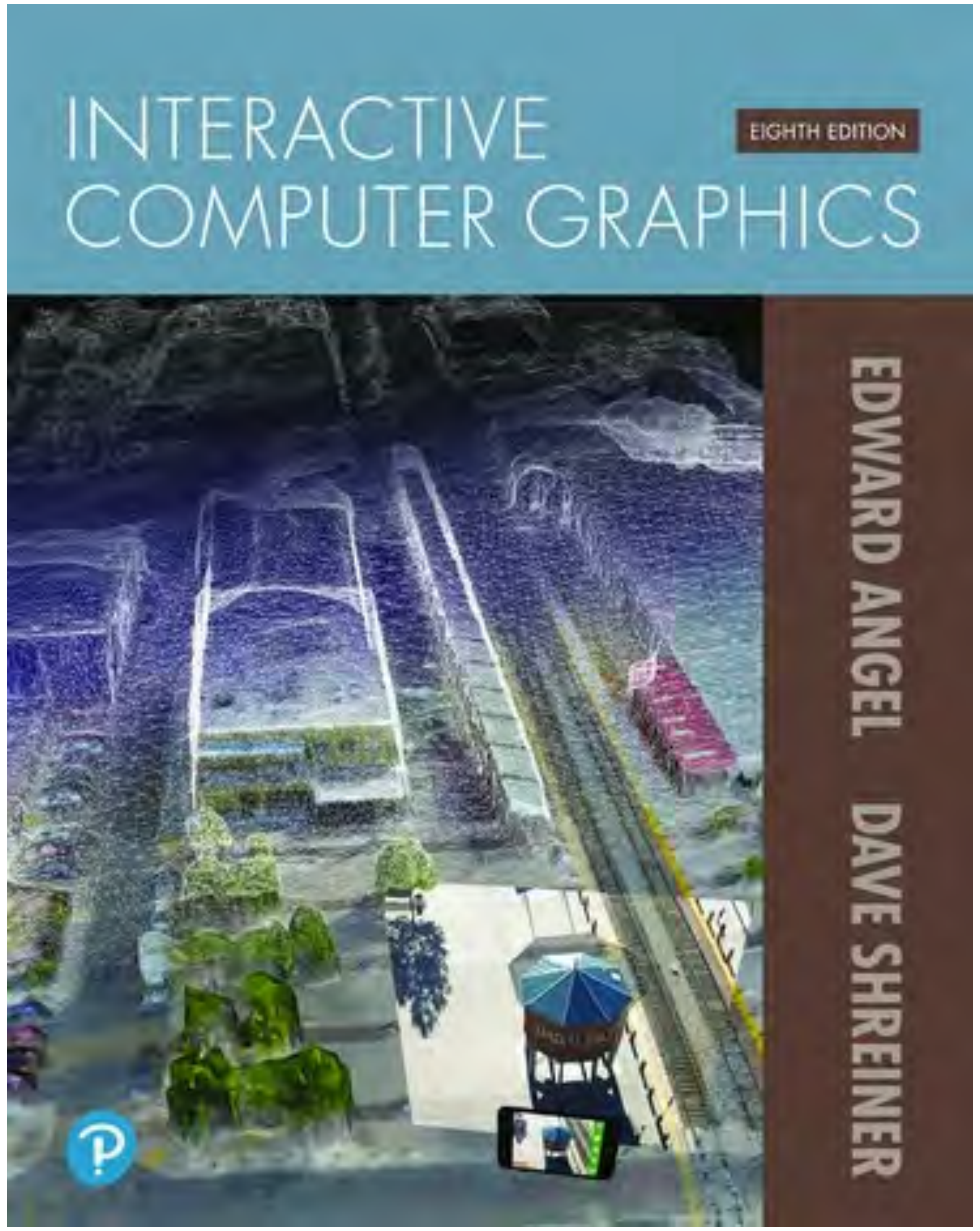
- Artistic/Creative asset creation
- Game design, shader effect design

Prerequisites & References

- Geometry
- Linear Algebra (Matrix multiplication)
- Prior high-level programming language experience (Course projects will be in JavaScript)
- Get familiar with basic usage of WebGL (WebGL Fundamentals Tutorials) by the 3rd week.
- (Optional) Interactive Computer Graphics: A Top-Down Approach with WebGL, 8th edition by Edward Angel, Dave Shreiner



Course Textbook and the Primary Author



Prof. Edward Angel
https://www.cs.unm.edu/~angel/

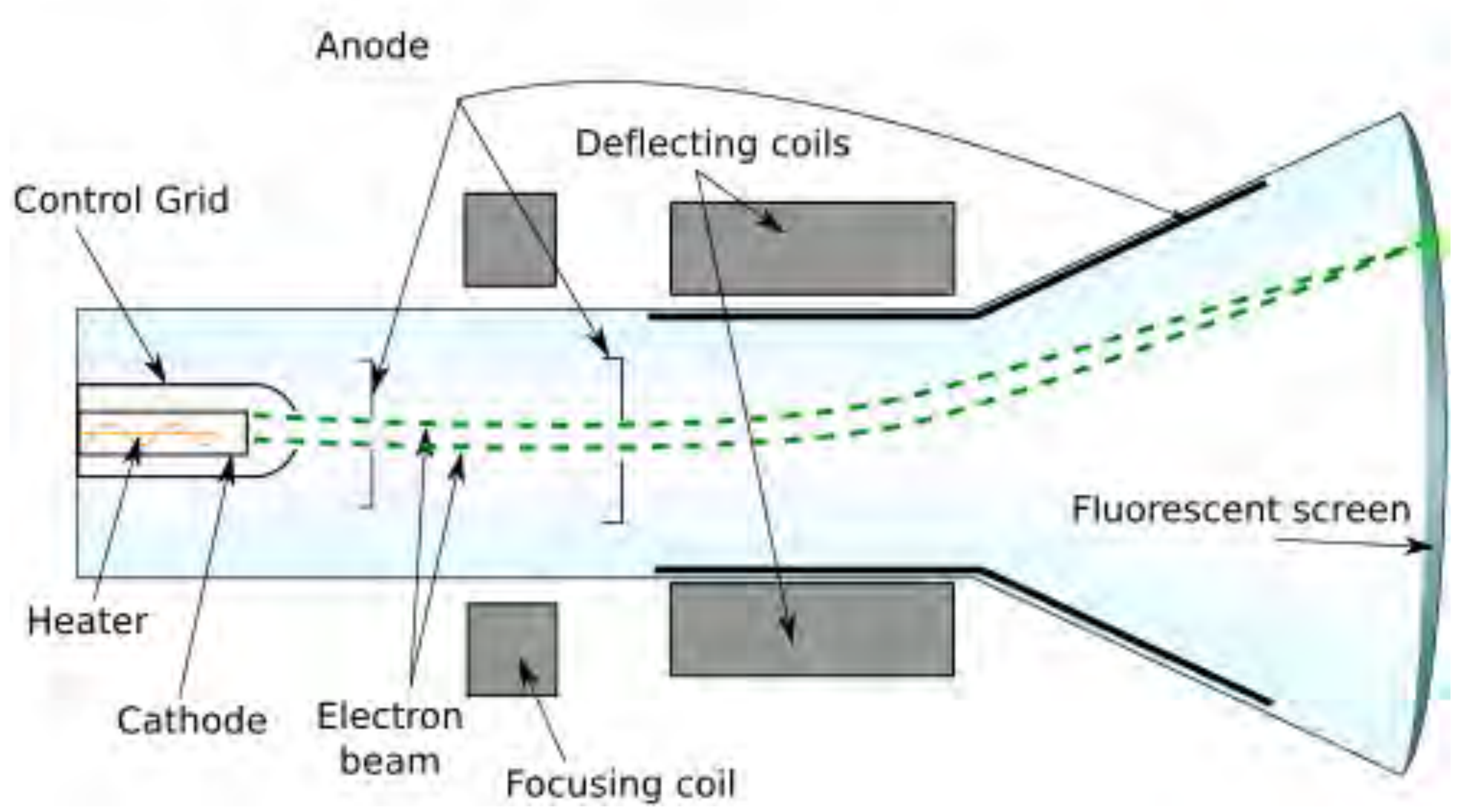
Grading

- 8 project assignments (60%)
  - Apply what you learned in class to creative content of your own choosing
  - results will look different and unique
  - Some projects will build on previous ones
  - Late penalty loses 1 point per day
- 1 midterm (15%)
- 1 final project (25%)

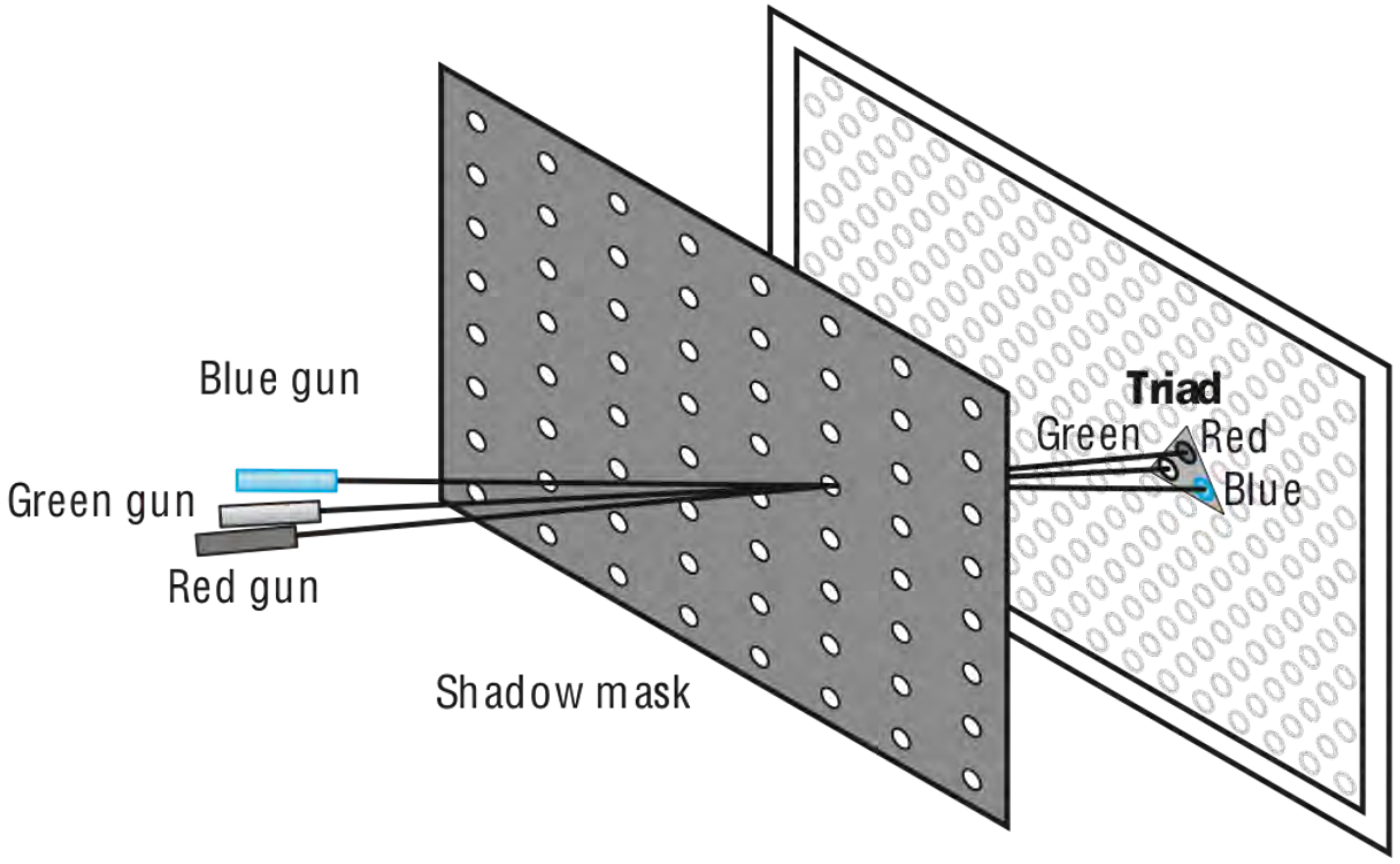
History

Some important Moments in CG History

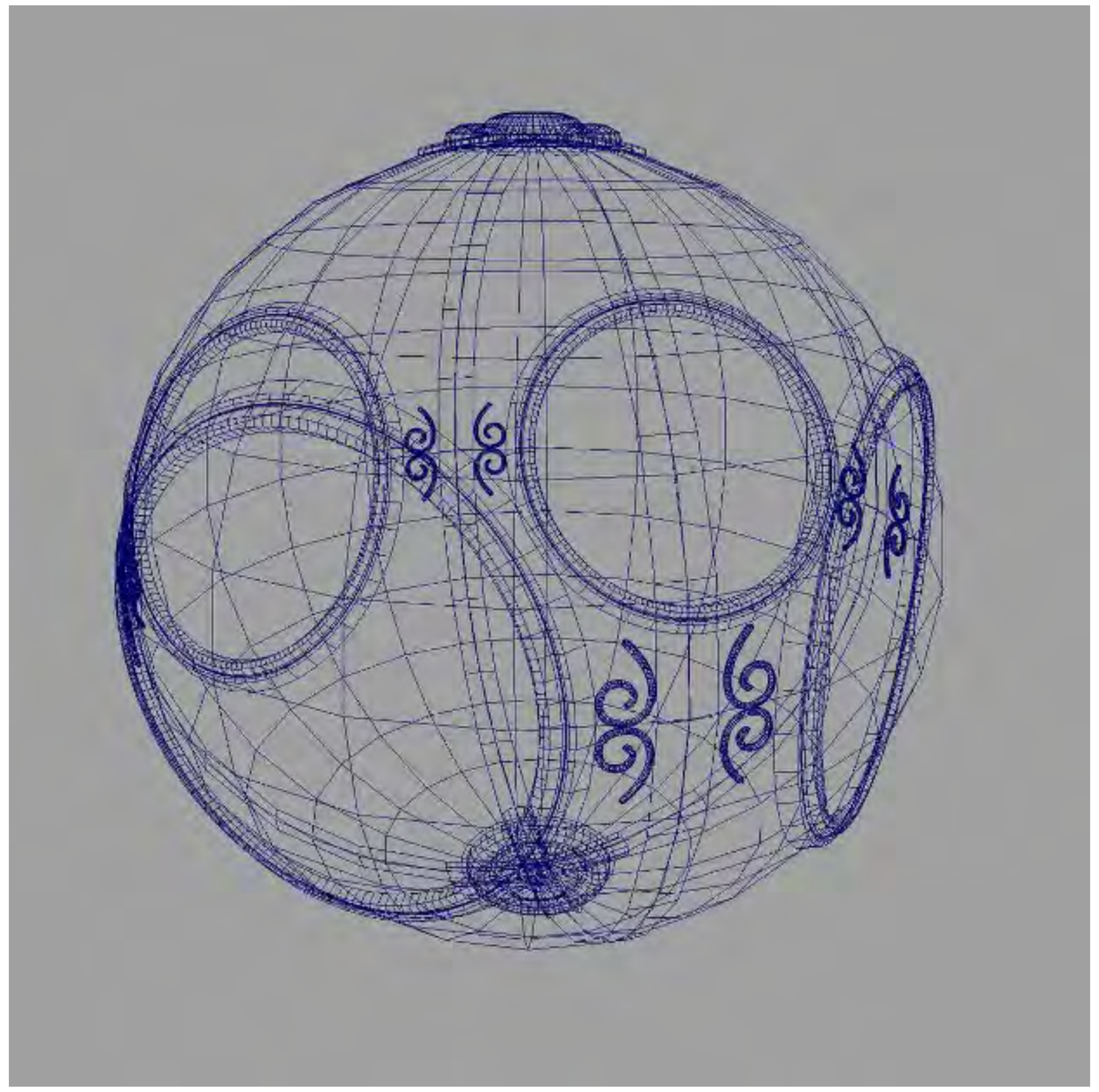
Cathode-ray tube (CRT), 1950s



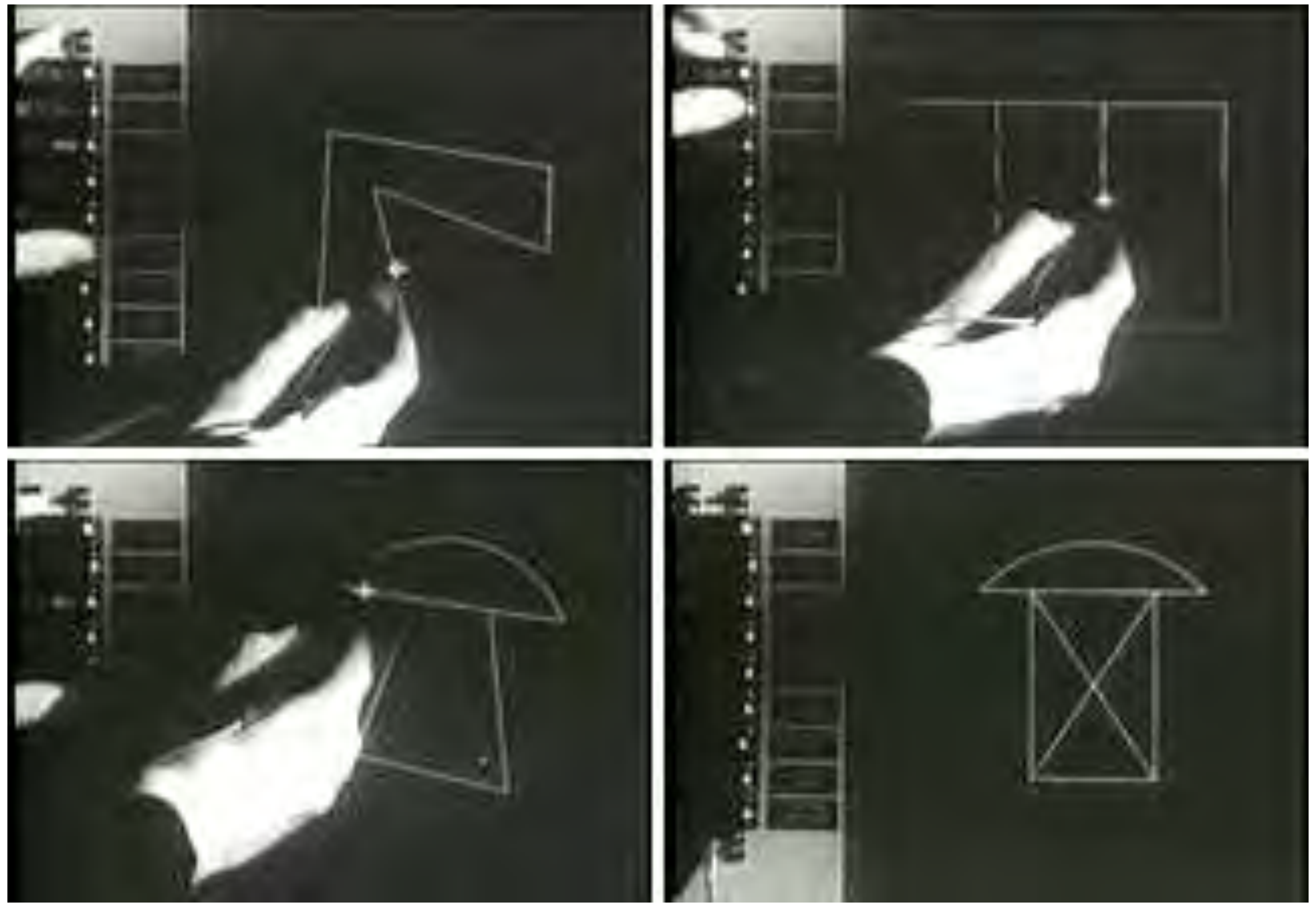
[source: Wikipedia]



1960s

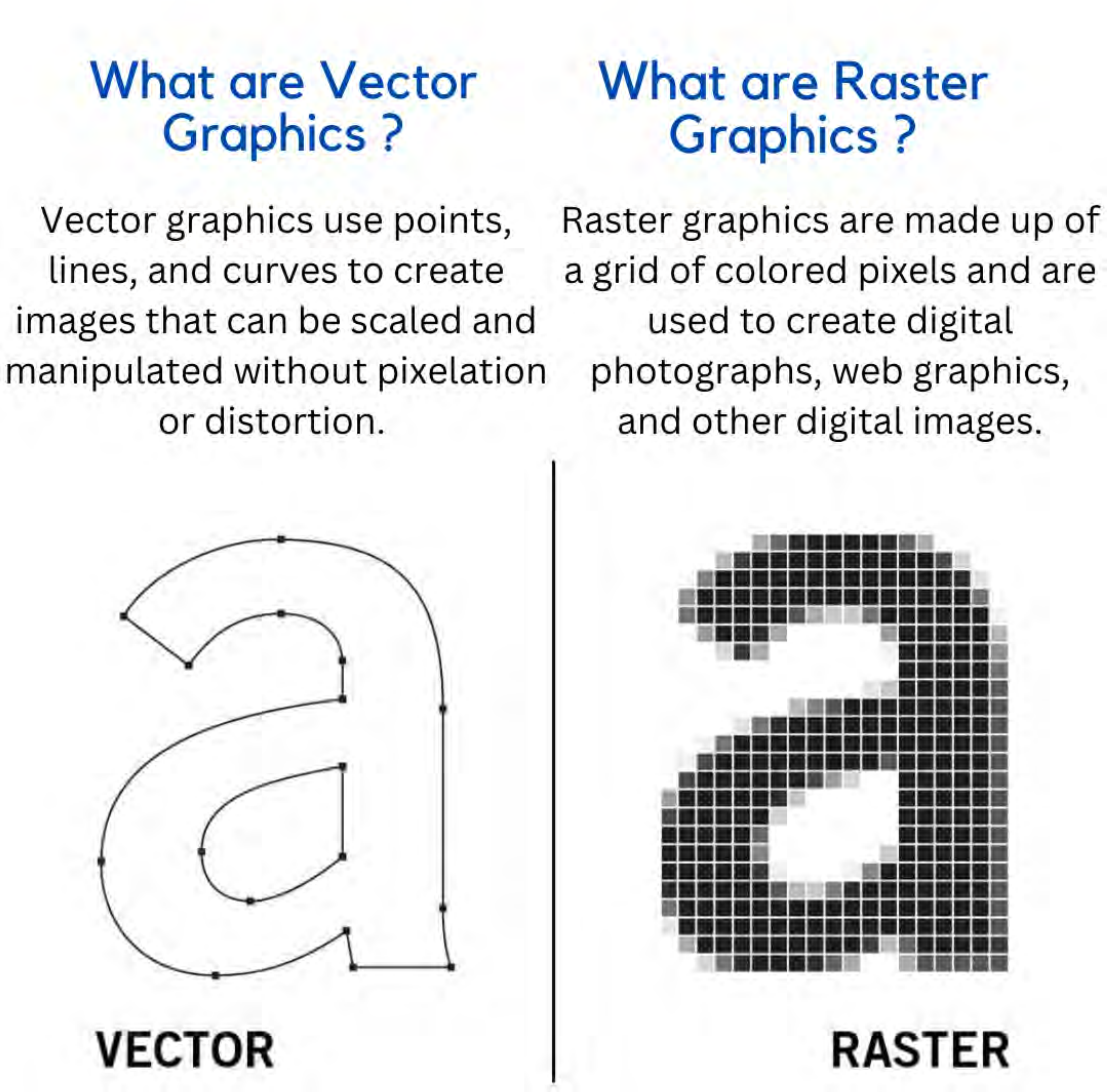


Wireframe graphics



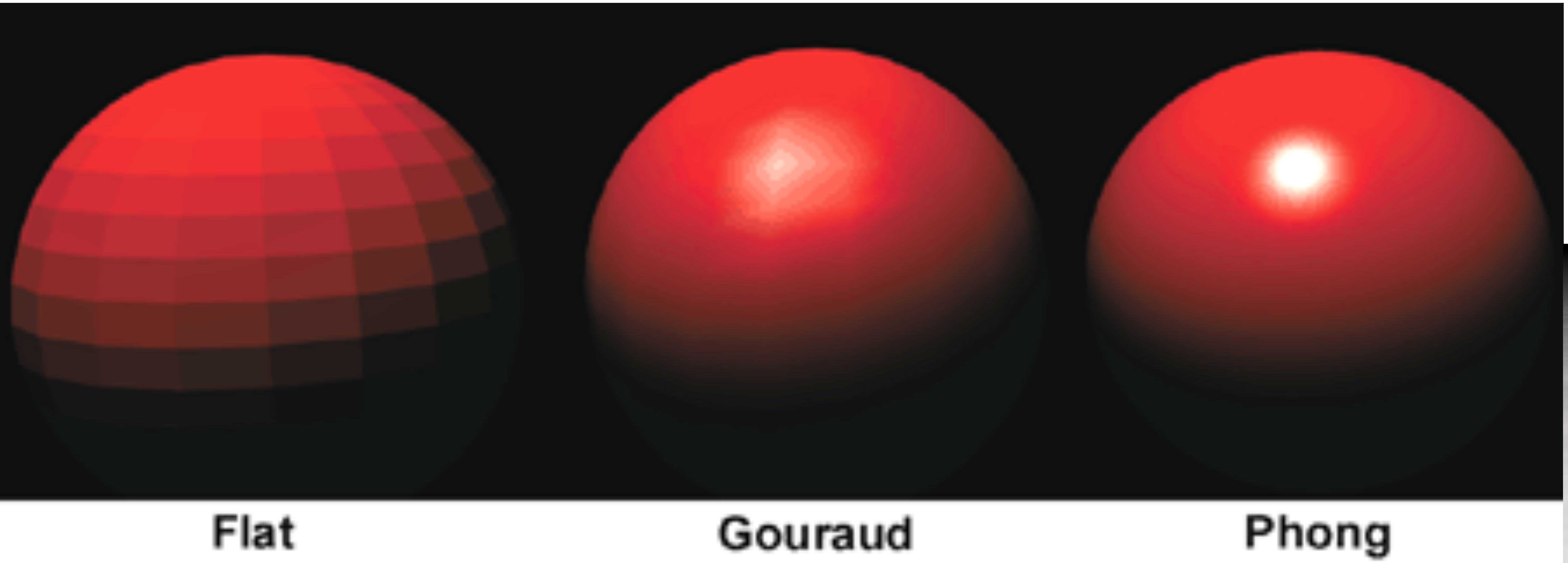
Sketchpad

1970s
Raster Graphics



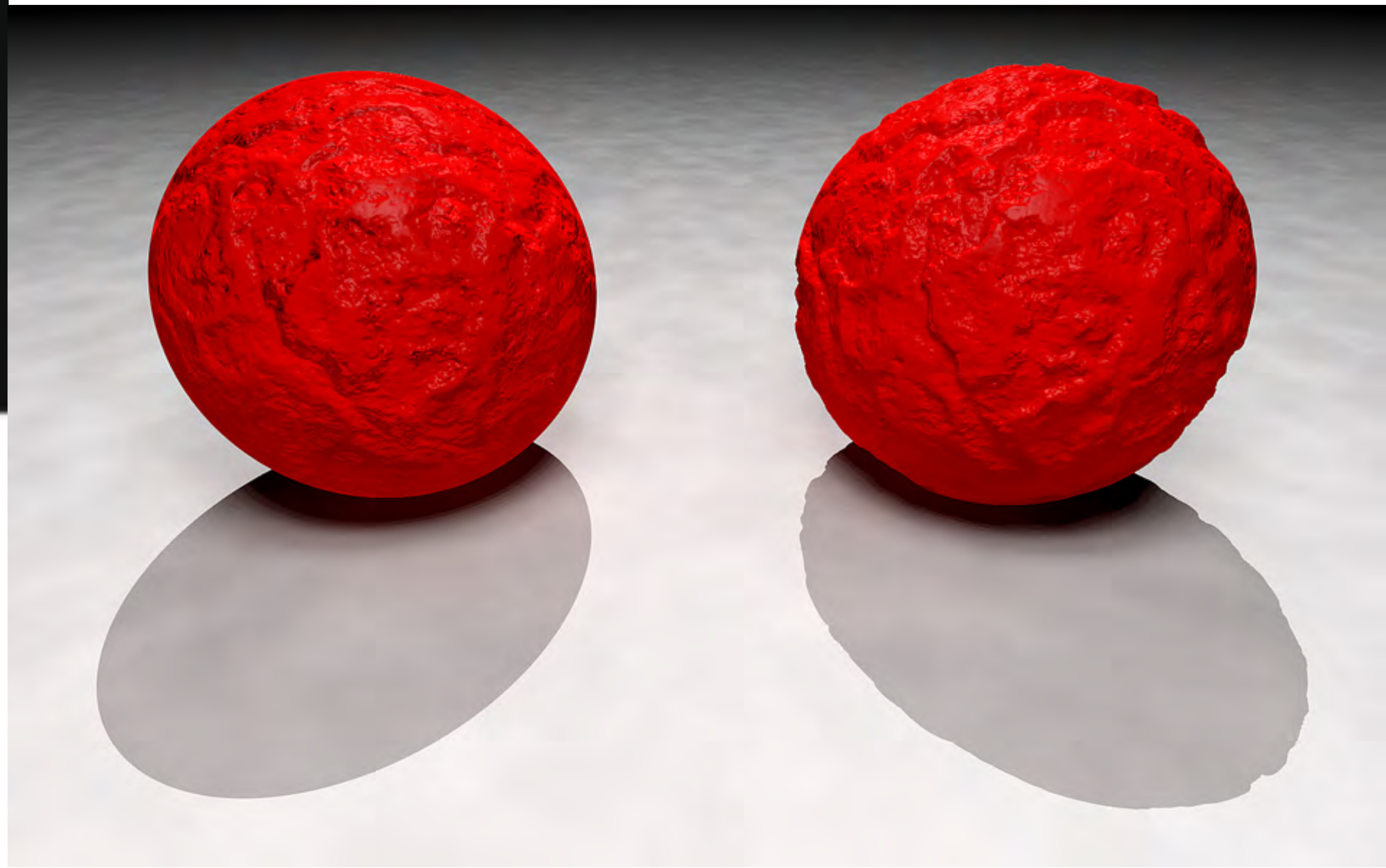
[source:devianart]

1970s
Realistic Rendering



[source:PCMag]

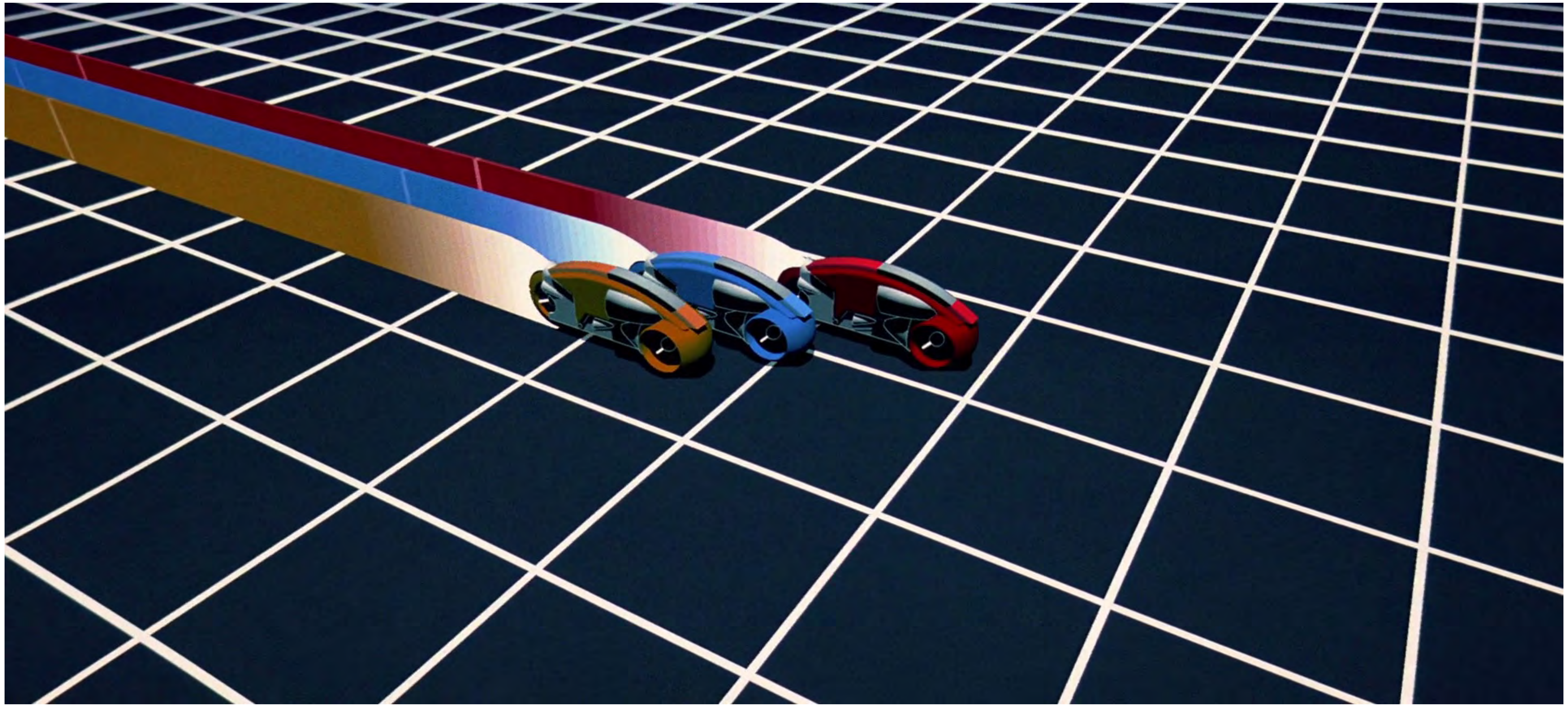
[source:Wikipedia]



[source:Wikipedia]

1980s

CG Movies and Video Games



[source:Tron]

[source:Pac-man]

[source:devianart]

1990s

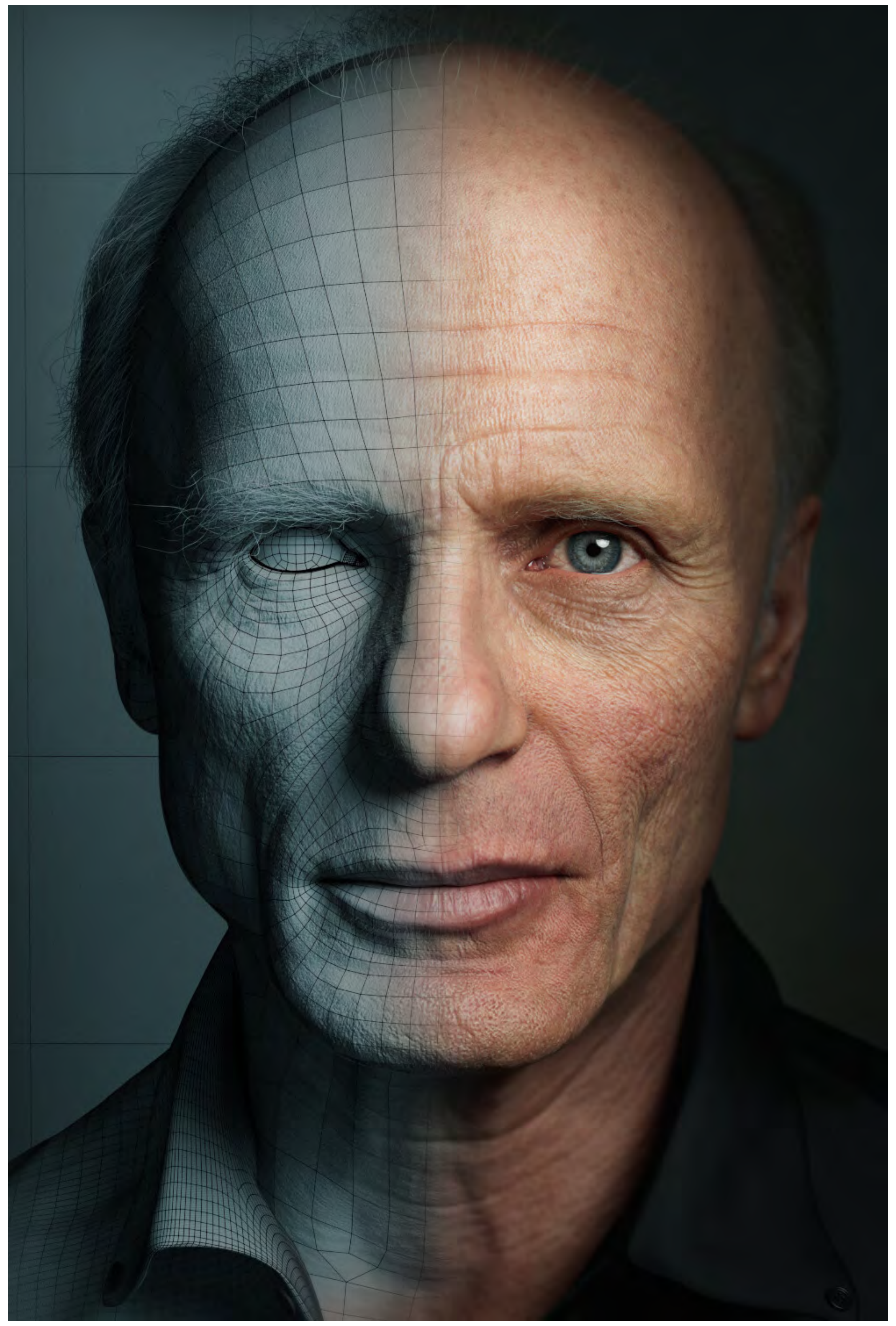
[source:OpenGL]

[source:Toy Story]

[source:Maya, Blender, Adobe]

2000s

[source:Lord Of the Rings]



[source:The Gnomon Workshop]

[source:Nvidia GeForce 256]

2010s - Now

[source:Qualcomm]

[source:WebGL]

[source:Oculus Rift]

Imagining What's Next …